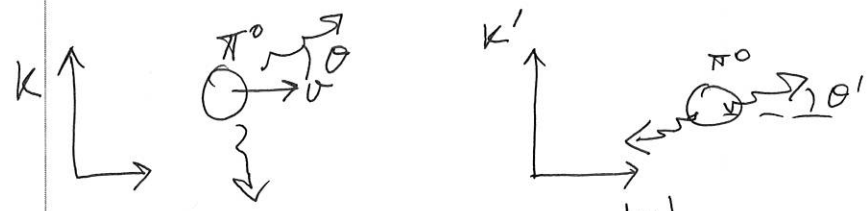
P1



$$\frac{dN}{d\Omega} = ? \qquad \frac{dN'}{d\Omega'} = \frac{1}{4\pi}$$

in K': $d\Omega' = d\varphi' \sin\theta' \, d\theta' = -d\varphi' \, d(\cos\theta')$

It is enough to consider one of the two photons in which π^0 decays.

Its 4-momentum is $(p')^\alpha = \left(\frac{\varepsilon'}{c}, p'\cos\theta', p'\sin\theta', 0\right)$

assuming its trajectory is in x-y plane.

$$\begin{pmatrix} \frac{\varepsilon}{c} \\ p_x \end{pmatrix} = \gamma \begin{pmatrix} 1 & \beta \\ \beta & 1 \end{pmatrix} \begin{pmatrix} \frac{\varepsilon'}{c} \\ p'_x \end{pmatrix} \quad \text{and} \quad p_y = p'_y$$

For a photon: $\varepsilon = \hbar\omega = \hbar c k$, $p = \hbar k$

$$\Rightarrow \begin{pmatrix} \hbar k \\ \hbar k \cos\theta \end{pmatrix} = \gamma \begin{pmatrix} 1 & \beta \\ \beta & 1 \end{pmatrix} \begin{pmatrix} \hbar k' \\ \hbar k' \cos\theta' \end{pmatrix}$$

$$\hbar k \sin\theta = \hbar k' \sin\theta'$$

OR
$$\begin{pmatrix} \hbar k' \\ \hbar k' \cos\theta' \end{pmatrix} = \gamma \begin{pmatrix} 1 & -\beta \\ -\beta & 1 \end{pmatrix} \begin{pmatrix} \hbar k \\ \hbar k \cos\theta \end{pmatrix}$$

$$\begin{cases} k' = \gamma(k - k\beta\cos\theta) \\ k'\cos\theta' = \gamma(k\cos\theta - \beta k) \end{cases}$$

$$\cos\theta' = \frac{k'\cos\theta'}{k'} = \frac{\gamma k(\cos\theta - \beta)}{\gamma k(1 - \beta\cos\theta)} = \frac{\cos\theta - \beta}{1 - \beta\cos\theta}$$

where $\beta = \frac{v}{c}$, v = velocity of π^0 along x.

Angle φ is perpendicular to π^0 axis of motion

$\Rightarrow$ it does not change during Lorentz transformation.

$\Rightarrow d\varphi = d\varphi'$.

$$d(\cos\theta') = d\left(\frac{\cos\theta - \beta}{1-\beta\cos\theta}\right) = \frac{d(\cos\theta)}{1-\beta\cos\theta} - \frac{(\cos\theta-\beta)(-\beta\, d\cos\theta)}{(1-\beta\cos\theta)^2} =$$

$$= \frac{d(\cos\theta)}{1-\beta\cos\theta}\left(1 + \frac{\beta(\cos\theta-\beta)}{1-\beta\cos\theta}\right) = \frac{1-\beta^2}{(1-\beta\cos\theta)^2}\, d(\cos\theta)$$

$$\Rightarrow d\Omega' = -d\varphi'\, d\cos\theta' = \frac{1-\beta^2}{(1-\beta\cos\theta)^2}\underbrace{\sin\theta\, d\theta\, d\varphi}_{d\Omega} = \frac{1-\beta^2}{(1-\beta\cos\theta)^2}\, d\Omega$$

Moreover, the number of γ-quanta is invariant: $dN = dN'$

$$\Rightarrow \frac{dN'}{d\Omega'} = \frac{dN}{d\Omega}\,\frac{(1-\beta\cos\theta)^2}{1-\beta^2} = \frac{1}{4\pi}$$

$$\Rightarrow \frac{dN}{d\Omega} = \frac{1-\beta^2}{4\pi(1-\beta\cos\theta)^2} = \frac{1}{4\pi\gamma^2(1-\beta\cos\theta)^2}$$

(P2) Number of γ-quanta emitted forward $= N_1$

$$N_1 = \int_0^{2\pi} d\varphi \int_0^{\pi/2} d\theta \sin\theta\, \frac{1}{4\pi\gamma^2(1-\beta\cos\theta)^2} = \frac{2\pi}{4\pi\gamma^2}\int_0^1 \frac{d(\cos\theta)}{(1-\beta\cos\theta)^2}$$

$$= \frac{1}{2\gamma^2\beta}\int d\left(\frac{1}{1-\beta\cos\theta}\right) = \frac{1}{2\beta\gamma^2}\,\frac{1}{1-\beta\cos\theta}\Big|_0^1 =$$

$$= \frac{1}{2\beta\gamma^2}\left(\frac{1}{1-\beta} - 1\right) = \frac{1}{2\beta\gamma^2}\,\frac{\beta}{1-\beta} = \frac{1}{2}\,\frac{1-\beta^2}{1-\beta} = \frac{1}{2}(1+\beta)$$

Number of γ-quanta emitted backward $= N_2$

$$N_2 = \int_0^{2\pi} d\varphi \int_{\pi/2}^{\pi} d\theta \sin\theta\, \frac{1}{4\pi\gamma^2(1-\beta\cos\theta)^2}$$

$$N_2 = \frac{1}{2\gamma^2}\int_{-1}^{0}\frac{d(\cos\theta)}{(1-\beta\cos\theta)^2} = \frac{1}{2\beta\gamma^2}\left.\frac{1}{1-\beta\cos\theta}\right|_{-1}^{0} =$$

$$= \frac{1}{2\beta\gamma^2}\left(1-\frac{1}{1+\beta}\right) = \frac{\beta}{2\beta\gamma^2(1+\beta)} = \frac{1}{2}\,\frac{1-\beta^2}{1+\beta} = \frac{1}{2}(1-\beta)$$

$$\Rightarrow f = \frac{N_1}{N_2} = \frac{1+\beta}{1-\beta}$$

Energy of π^0 meson is $\mathcal{E}_\pi = \dfrac{m_\pi c^2}{\sqrt{1-\beta^2}}$

But $f = \dfrac{1+\beta}{1-\beta} \Rightarrow (1-\beta)f = 1+\beta \Rightarrow \beta = \dfrac{f-1}{f+1}$

so that $1-\beta^2 = 1-\left(\dfrac{f-1}{f+1}\right)^2 = \dfrac{4f}{(f+1)^2}$

$$\Rightarrow \mathcal{E}_\pi = m_\pi c^2\,\frac{1}{\sqrt{\dfrac{4f}{(f+1)^2}}} = \frac{1}{2}\,m_\pi c^2\,\frac{f+1}{\sqrt{f}}.$$

(P3) Photon in K' $(p')^\alpha = \left(\dfrac{\mathcal{E}'}{c}, p'_x, p'_y, 0\right)$ $\mathcal{E}' = \frac{1}{2}m_\pi c^2$.

in K $p^\alpha = \left(\dfrac{\mathcal{E}}{c}, p_x, p_y, 0\right)$

$$\begin{pmatrix}\mathcal{E}/c\\ p_x\end{pmatrix} = \gamma\begin{pmatrix}1 & \beta\\ \beta & 1\end{pmatrix}\begin{pmatrix}\mathcal{E}'/c\\ p'_x\end{pmatrix} \Rightarrow \mathcal{E} = \gamma(\mathcal{E}' + c\beta p'_x)$$

$$p'_x = p'\cos\theta' = \frac{\mathcal{E}'}{c}\cos\theta'$$

$$\mathcal{E} = \gamma\left(\mathcal{E}' + v\,\frac{\mathcal{E}'}{c}\cos\theta'\right) = \gamma\mathcal{E}'(1+\beta\cos\theta')$$

But from (P1) $\cos\theta' = \dfrac{\cos\theta-\beta}{1-\beta\cos\theta}$

$$\Rightarrow 1+\beta\cos\theta' = 1+\frac{\beta(\cos\theta-\beta)}{1-\beta\cos\theta} = \frac{1-\beta^2}{1-\beta\cos\theta}$$

$$\Rightarrow \mathcal{E} = \gamma \mathcal{E}' \frac{1-\beta^2}{1-\beta\cos\theta} = \frac{\mathcal{E}'}{\gamma(1-\beta\cos\theta)}$$

Thus $\mathcal{E}_{max} = \dfrac{\mathcal{E}'}{\gamma(1-\beta)}$ (for $\theta = 0$) $= \mathcal{E}'\sqrt{\dfrac{1+\beta}{1-\beta}}$

$\mathcal{E}_{min} = \dfrac{\mathcal{E}'}{\gamma(1+\beta)}$ (for $\theta = \pi$) $= \mathcal{E}'\sqrt{\dfrac{1-\beta}{1+\beta}}$

$$\frac{\mathcal{E}'}{\gamma(1+\beta)} \leq \mathcal{E} \leq \frac{\mathcal{E}'}{\gamma(1-\beta)} \qquad \mathcal{E}' = \tfrac{1}{2} m_\pi c^2, \ \beta = \frac{v}{c}, \ \gamma = \frac{1}{\sqrt{1-\beta^2}}$$

Another interesting consequence of this is:

$$d\mathcal{E} = \frac{\mathcal{E}'}{\gamma} d\left(\frac{1}{1-\beta\cos\theta}\right) = \frac{\mathcal{E}'}{\gamma} \frac{\beta \, d(\cos\theta)}{(1-\beta\cos\theta)^2}$$

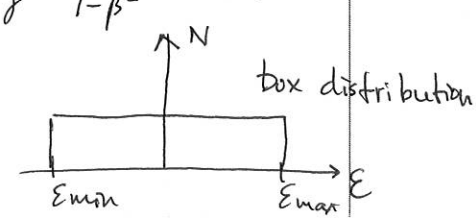
But $d\Omega = 2\pi \sin\theta \, d\theta = -2\pi \, d(\cos\theta)$ since nothing depends on φ.

$$d\mathcal{E} = -\frac{\mathcal{E}'}{2\pi\gamma} \frac{\beta}{(1-\beta\cos\theta)^2} d\Omega$$

From P1 $\quad dN = \dfrac{1}{4\pi\gamma^2(1-\beta\cos\theta)^2} d\Omega$

$$\Rightarrow \frac{dN}{d\mathcal{E}} = \frac{1}{4\pi\gamma^2(1-\beta\cos\theta)^2} \frac{-2\pi\gamma(1-\beta\cos\theta)^2}{\mathcal{E}'\beta} = -\frac{1}{2\beta\gamma\mathcal{E}'}$$

Distribution of photon energies is constant.

Since $\mathcal{E}_{max} - \mathcal{E}_{min} = \dfrac{\mathcal{E}'}{\gamma}\left(\dfrac{1}{1-\beta} - \dfrac{1}{1+\beta}\right) = \dfrac{\mathcal{E}'}{\gamma} \dfrac{2\beta}{1-\beta^2} = 2\beta\gamma\mathcal{E}'$

We have $\quad dN = \dfrac{|d\mathcal{E}|}{\mathcal{E}_{max} - \mathcal{E}_{min}}$